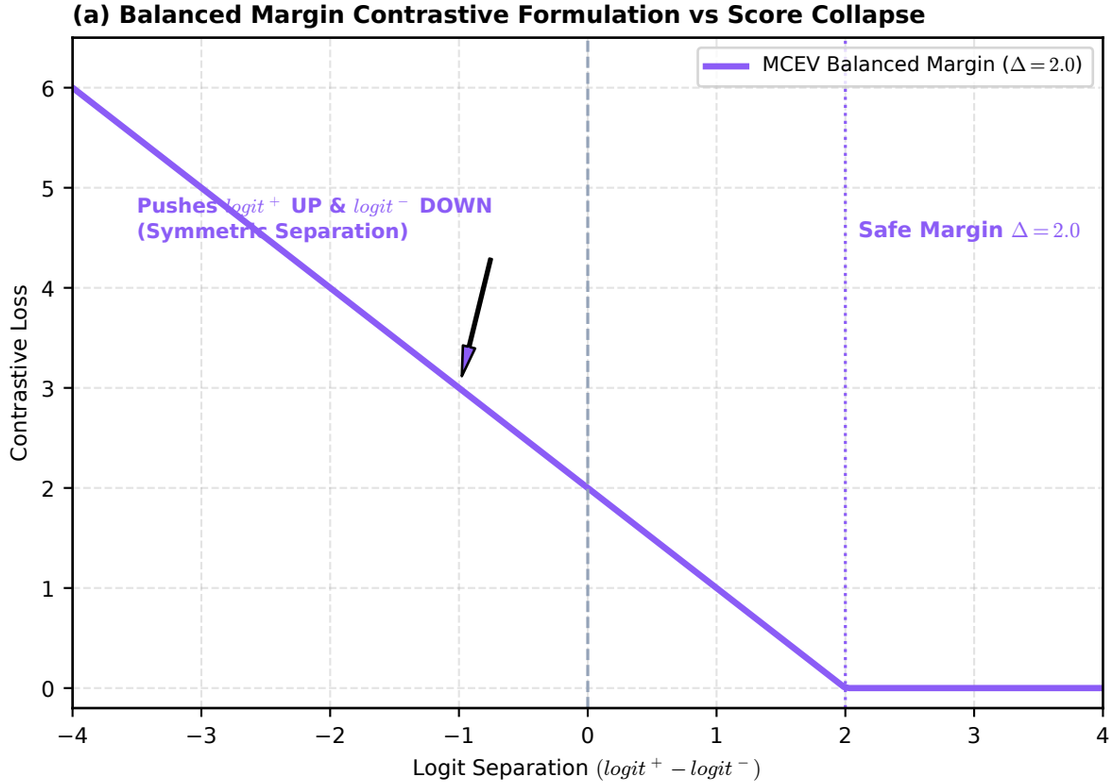
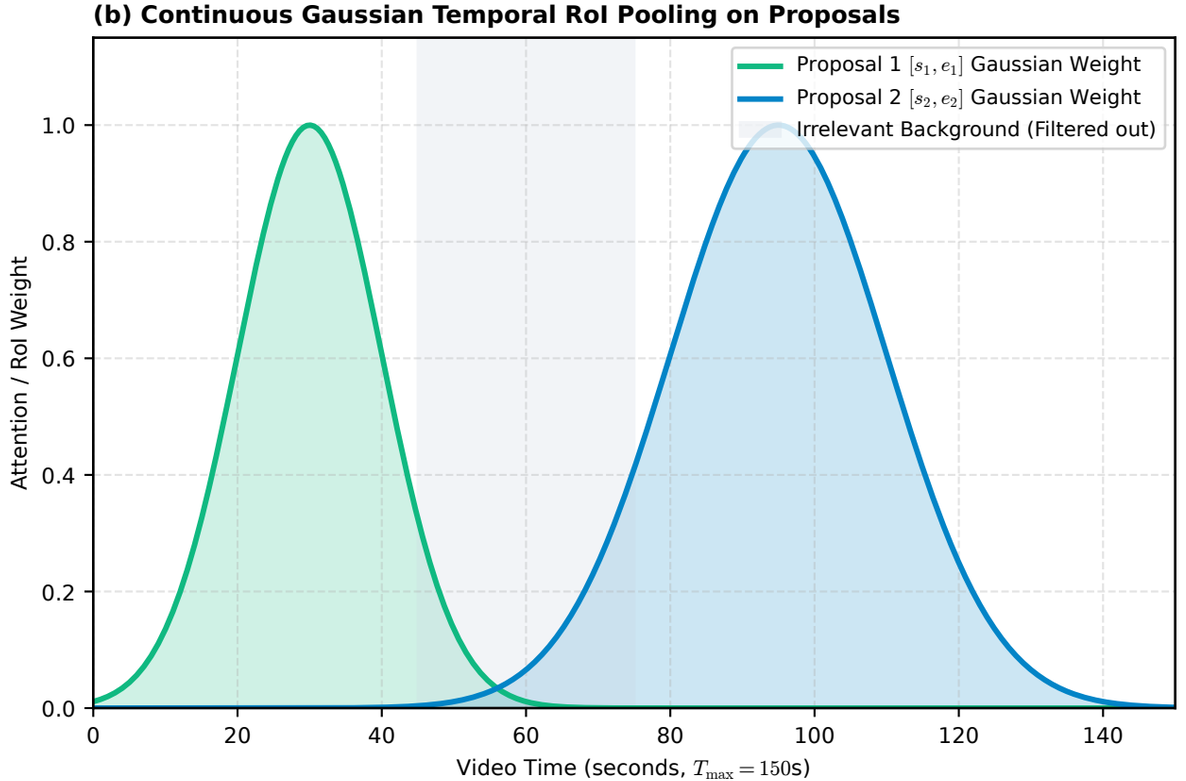

(a) Balanced Margin Contrastive Formulation vs Score Collapse
MCEV Balanced Margin ($\Delta = 2.0$)
Pushes $logit^+$ UP & $logit^-$ DOWN
(Symmetric Separation)
Safe Margin $\Delta = 2.0$
Contrastive Loss
Logit Separation ($logit^+ - logit^-$)
0
1
2
3
4
5
6
−4
−3
−2
−1
0
1
2
3
4


(b) Continuous Gaussian Temporal RoI Pooling on Proposals
Proposal 1 $[s_1, e_1]$ Gaussian Weight
Proposal 2 $[s_2, e_2]$ Gaussian Weight
Irrelevant Background (Filtered out)
Attention / RoI Weight
Video Time (seconds, $T_{\max} = 150$s)
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
120
140